

# Finite Resonance Control of Nonlinear Hahn–Dulac Transseries

Exact logarithmic slopes, analytic realization,  
and accumulation thresholds

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## Abstract

We study the nonlinear regular-singular system  $(x d/dx - A)y = F(x, y)$ , where  $A$  has real spectrum and the logarithm-free coefficients of  $F$  have support in a positive real Hahn semigroup. The support may have finite accumulation points. Every choice of the allowed resonant constants determines a unique formal solution  $y = \sum_{\gamma > 0} x^\gamma P_\gamma(\log x)$ . Although its logarithmic degrees can be unbounded, their maximal ratio to the exponent is determined by finitely many resonant polynomial blocks. The determining polynomial blocks depend polynomially on only finitely many input coordinates. The slope sublevel sets are algebraic, and the maximum is invariant under invertible logarithm-free local changes of the dependent variables.

We also prove a universal analytic-realization criterion: every absolutely convergent input has a convergent normalized Hahn–Dulac solution precisely when the nonresonant semigroup exponents stay uniformly separated from the eigenvalues of  $A$ . A weighted polynomial norm supplies explicit contraction and remainder certificates. Failure of the gap condition admits arbitrarily small, logarithm-free counterexamples.

An explicit nonlinear Riccati family with actions  $2 - 1/n$  separates finite resonance control from convergence particularly sharply. Its logarithmic slope depends on the single polynomial  $b + ca^2$ , whereas absolute convergence is equivalent to  $\sum n|t_n| < \infty$ . Thus coefficients  $t_n = \varepsilon n^{-p}$  have the exact threshold  $p > 2$ , even in the presence of nonlinear feedback. Complete conventional proofs, finite symbolic verification, and twelve further research directions are provided. Classical Dulac majorants and the repository’s linear resonance results are acknowledged; historical priority is not asserted.

**Scope.** Positive real actions; one logarithm; first-order systems with no derivatives inside  $F$ . These are human-readable proofs, not a Lean formalization or an independently peer-reviewed report.

**Inspected repository snapshot.** 3d5973524506411392a911470b5ddc35521568ea

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# 1 The problem, the contribution, and its boundaries

Two distinct kinds of finiteness occur in transseries calculations. A coefficient can have only finitely many algebraic ancestors even when infinitely many exponents lie below it. Conversely, finitely many resonances can generate infinitely many nonzero logarithmic powers once nonlinear multiplication is allowed. Confusing these two facts leads to an inappropriate choice of ambient space and, more seriously, to incorrect inferences about convergence.

This article asks a concrete extension question:

For a nonlinear regular-singular system with well-ordered real actions, can all logarithmic growth be determined from finite resonance data, and when does the resulting formal solution define an actual convergent function?

The answer has three parts. The correct global invariant is a *degree-to-action slope*, not a finite bound on logarithmic degree. That slope is determined by a finite algebraic certificate. Convergence, however, can depend on infinitely many coefficients that do not occur in that certificate.

## 1.1 Relation to the inspected ProveIt materials

The repository’s support module contains additive versions of Neumann’s support and finite-decomposition lemmas, obtained through explicit Mathlib bridges [1]. The finite-divisor argument used below is a direct application of that established infrastructure. The residue-obstruction package treats finite obstructions and logarithmic-depth promotion for linear operators [2]. The Hahn–Fuchsian package gives a finite resonance matrix for a linear system, a logarithmic-degree filtration, and a sharp universal convergence criterion for its normalized gauge [3].

We do not rebrand those linear results as nonlinear discoveries. Here there is generally no finite-dimensional vector space of solutions and no single nilpotent matrix whose nilpotency index bounds every logarithmic degree. Indeed, the scalar equation in Section 9 has coefficients of degree exactly  $n$  at action  $n$ . The replacement proved here is an exact finite-resonance formula for the maximal slope, together with nonlinear algebraic cancellation loci and invariance under local coordinate changes. Our analytic gap measures distances to *eigenvalues*, whereas the linear gauge-normalization problem naturally involves *eigenvalue differences*. They are different operators.

## 1.2 Established ingredients and proposed additions

General fixed-point methods for generalized series are established; van der Hoeven proves an implicit-function theorem for Noetherian operators [4]. Our elementary valuation construction is a special-purpose instance, not a new general implicit-function theorem. Gontsov and Goryuchkina prove a majorant convergence theorem for generalized power series [5].

In their Dulac-series paper, Goryuchkina and Gontsov already obtain a linear degree bound in the integer action and use coefficient norms, rescaling of the logarithmic variable, and bounds for polynomial Euler operators [6, Lemmas 2–3]. Neither the existence of a linear degree bound nor that norm method is claimed as new here. We adapt those ingredients to finite resonances and possibly non-locally-finite Hahn supports.

The proposed additions are the exact maximum formula of Theorem 4.4, its finite algebraic certificate and coordinate-invariance refinements, the nonlinear formulation of the universal

gap equivalence, and the input-specific nonlinear accumulation theorem of Section 10. These statements are proved below. The literature and repository comparison was targeted, not exhaustive; independent assessment of historical novelty is still needed. In particular, the article does not claim to settle a named classical conjecture or all transseries differential equations.

### 1.3 A compact statement of the main results

Let  $\Gamma \subset [0, \infty)$  be a nontrivial well-ordered additive submonoid, with  $\delta = \min \Gamma_{>0} > 0$ , and let  $A$  be a constant complex matrix with real spectrum. Consider

$$(D - A)y = F(x, y), \quad D = x \frac{d}{dx}, \quad F(x, z) = \sum_{\alpha, k} f_{\alpha, k} x^\alpha z^k, \quad (1.1)$$

where  $\alpha \in \Gamma$ ,  $k \in \mathbb{N}^m$ , and  $f_{0,0} = f_{0,e_j} = 0$ . Set  $\mathcal{R} = \text{spec}(A) \cap \Gamma_{>0}$ . For a polynomial vector  $P$ , write  $\ell(P) = \max(0, \deg P)$ , with  $\ell(0) = 0$ . The normalized formal solution satisfies

$$\Theta(y) := \sup_{\gamma \in \Gamma_{>0}} \frac{\ell(P_\gamma)}{\gamma} = \max \left( \{0\} \cup \left\{ \frac{\ell(P_\lambda)}{\lambda} : \lambda \in \mathcal{R} \right\} \right). \quad (1.2)$$

Only actions in the finite set

$$\mathcal{D} = \bigcup_{\lambda \in \mathcal{R}} \{\alpha \in \Gamma : \lambda - \alpha \in \Gamma\} \quad (1.3)$$

are needed to calculate the right-hand side. The exact slope is unchanged by any logarithm-free local transformation with invertible constant linear part.

For convergence, the additional universal condition is

$$g(\Gamma, A) := \inf_{\substack{\gamma \in \Gamma_{>0}, \\ \lambda \in \text{spec}(A) \\ \gamma \neq \lambda}} |\gamma - \lambda| > 0. \quad (1.4)$$

When it holds, every absolutely convergent  $F$  has an absolutely convergent normalized solution for every allowed collection of resonant constants. When it fails, even logarithm-free linear forcing supplies arbitrarily small counterexamples. The quantifier “every” is essential: a particular nonlinear equation can have a convergent solution despite  $g(\Gamma, A) = 0$ .

## 2 The Hahn–Dulac algebra and finite divisors

### 2.1 Orientation, support, and polynomial blocks

The independent variable tends to zero, so smaller positive exponents are dominant. All well-orderings in this article use the usual order on  $\mathbb{R}$ . This is opposite to the descending-monomial convention commonly used at positive infinity.

Fix the monoid  $\Gamma$  from the introduction. We use formal sums

$$\mathcal{D}_\Gamma = \left\{ \sum_{\gamma \in \Gamma} x^\gamma P_\gamma(L) : P_\gamma \in \mathbb{C}[L] \right\}, \quad \mathfrak{m}_\Gamma = \{y \in \mathcal{D}_\Gamma : P_0 = 0\}. \quad (2.1)$$

The coefficients at individual actions are finite polynomials, but their degrees need not have a common bound. Thus  $\mathcal{D}_\Gamma$  is larger than the polynomial ring in  $L$  over an ordinary Hahn

coefficient ring. In the monomial ordering which first compares  $x$ -actions and then logarithmic powers, its support remains well based: the first nonempty action has only finitely many logarithmic terms.

Define a strong derivation by

$$D(x^\gamma P(L)) = x^\gamma(\gamma P(L) + P'(L)), \quad DL = 1. \quad (2.2)$$

In the analytic sections,  $L = \text{Log } x$  on a specified logarithmic cover. The outer valuation  $v(y)$  is the least action with nonzero polynomial coefficient, and  $v(0) = +\infty$ . For vectors, use the least action occurring in any component. This valuation ignores logarithmic degree.

**Lemma 2.1** (Finite decompositions and substitution). *For each  $\beta \in \Gamma$ , the set*

$$\{(\alpha_1, \dots, \alpha_n) \in \Gamma^n : \alpha_1 + \dots + \alpha_n = \beta\}$$

*is finite for every fixed  $n$ . If all entries are positive, then  $n \leq \lfloor \beta/\delta \rfloor$ . Consequently, products in  $\mathcal{D}_\Gamma$  and substitutions  $F(x, y)$  with  $y \in \mathbf{m}_\Gamma^m$  are coefficientwise well defined.*

*Proof.* For two summands this is Neumann’s finite-decomposition lemma. An elementary real-order proof is also useful: infinitely many distinct pairs with fixed sum would have a subsequence whose first coordinates strictly increase, forcing the second coordinates to strictly decrease, contrary to well-ordering. Inducting on  $n$  gives the general case. If every summand is positive, their sum is at least  $n\delta$ .

At action  $\beta$ , a monomial  $x^\alpha y^k$  can contribute only when  $|k| \leq \lfloor \beta/\delta \rfloor$ . There are finitely many such multi-indices; for each, all action decompositions are finite. This also allows  $\alpha = 0$ . Each contribution is a finite product of polynomials in  $L$ , so the resulting block is a polynomial.  $\square$

The same argument verifies the Leibniz rule for  $D$ : every coefficient comparison involves a finite sum. No interchange of analytic limits is involved at this stage.

**Definition 2.2** (Divisor set). For  $\beta \in \Gamma$ , define

$$\text{Div}_\Gamma(\beta) = \{\alpha \in \Gamma : \beta - \alpha \in \Gamma\}. \quad (2.3)$$

An action  $\alpha$  in this set is called a divisor action of  $\beta$ . This is additive divisibility, not divisibility of real numbers.

**Lemma 2.3** (Finite ancestry). *The set  $\text{Div}_\Gamma(\beta)$  is finite. If  $\alpha$  divides  $\beta$  and  $\eta$  divides  $\alpha$ , then  $\eta$  divides  $\beta$ . Every partial sum in a decomposition of  $\beta$  into monoid elements is a divisor action of  $\beta$ .*

*Proof.* The map  $\alpha \mapsto (\alpha, \beta - \alpha)$  is a bijection to the finite two-factor antidiagonal from Lemma 2.1. Transitivity follows from  $\beta - \eta = (\beta - \alpha) + (\alpha - \eta) \in \Gamma$ . The complementary partial sum proves the final assertion.  $\square$

*Remark 2.4* (Finite divisibility is not local finiteness). For  $E = \{2 - 1/n : n \geq 1\}$ , the generated monoid  $\Gamma = \langle E \rangle$  is well ordered, although  $\Gamma \cap [0, 2]$  is infinite. To see the well-ordering, each fixed finite sumset of  $E$  is well ordered by Neumann’s support lemma, and a bounded interval meets only finitely many possible word lengths, because  $\min E = 1$ . Nevertheless  $\text{Div}_\Gamma(2) = \{0, 1, 2\}$ . This example is the main reason to formulate certificates with divisor sets rather than with all actions below a cutoff.

### 3 The normalized nonlinear formal solution

#### 3.1 Equation and normalization data

Fix an integer  $m \geq 1$ , and write  $\mathbb{N} = \{0, 1, 2, \dots\}$ . Write the generalized eigenspace decomposition as

$$\mathbb{C}^m = \bigoplus_{\lambda \in \text{spec}(A)} V_\lambda, \quad A|_{V_\lambda} = \lambda I + N_\lambda, \quad N_\lambda^{s_\lambda} = 0,$$

where  $s_\lambda$  is the largest Jordan block size for  $\lambda$ . Let  $\Pi_\lambda$  denote the corresponding projection. The input is

$$(D - A)y = F(x, y), \quad F(x, z) = \sum_{\alpha \in \Gamma, k \in \mathbb{N}^m} f_{\alpha, k} x^\alpha z^k, \quad f_{0,0} = f_{0,e_j} = 0. \quad (3.1)$$

The notation  $z^k$  means  $z_1^{k_1} \cdots z_m^{k_m}$ . Every  $f_{\alpha, k}$  is a vector in  $\mathbb{C}^m$ . No convergence hypothesis is imposed yet.

The vanishing conditions have a precise role: constant forcing has positive action, and all linear terms of action zero have already been absorbed into  $A$ . Pure nonlinear terms at action zero remain allowed. For each  $\lambda \in \mathcal{R} = \text{spec}(A) \cap \Gamma_{>0}$ , choose  $c_\lambda \in V_\lambda$  and impose

$$\Pi_\lambda P_\lambda(0) = c_\lambda. \quad (3.2)$$

These are constants of the polynomial block at a resonant action; they are not values of the analytic solution at  $x = 0$ .

#### 3.2 The polynomial Euler solver

**Lemma 3.1** (One generalized eigenspace). *Let  $N^s = 0$ , let  $Q$  be a polynomial vector, and consider*

$$(aI - N + \partial_L)P = Q. \quad (3.3)$$

*If  $a \neq 0$ , put  $B = (aI - N)^{-1}$ . The unique polynomial solution is*

$$P = \sum_{j=0}^{\deg Q} (-1)^j B^{j+1} \partial_L^j Q, \quad (3.4)$$

*with the zero solution when  $Q = 0$ . A nonzero  $Q$  and its solution have the same degree.*

*If  $a = 0$ , the unique solution with  $P(0) = c$  is*

$$P = e^{NL}c + \sum_{j=0}^{s-1} N^j I_L^{j+1} Q, \quad I_L Q(L) = \int_0^L Q(t) dt. \quad (3.5)$$

*In this case  $\ell(P) \leq \max(s-1, \ell(Q) + s)$ .*

*Proof.* When  $a \neq 0$ , the inverse of  $aI - N$  exists by nilpotence. The operators  $B$  and  $\partial_L$  commute, and differentiation is nilpotent on every bounded-degree polynomial space. Multiplying the finite geometric sum in (3.4) by  $(aI - N)(I + B\partial_L)$  gives  $Q$ . Its leading coefficient is the invertible image under  $B$  of the leading coefficient of  $Q$ , so the degree is unchanged. A nonzero polynomial in the kernel would have a highest coefficient annihilated by  $aI - N$ , which is impossible.

At resonance, the homogeneous solution with constant term  $c$  is  $e^{NL}c$ , a finite polynomial. Differentiating the sum in (3.5) and subtracting its product by  $N$  telescopes: all terms cancel

except  $Q$ , because  $N^s = 0$ . Every integral in that sum vanishes at zero. The difference of two normalized solutions solves  $P' = NP$  with zero constant term, and its successive coefficients are therefore zero. Integration raises degree by one and at most  $s$  integrations occur, proving the bound.  $\square$

Apply this lemma at action  $\gamma$  with  $a = \gamma - \lambda$  on each generalized eigenspace. Let  $S$  be the resulting right inverse of  $D - A$ , with zero resonant constants. It does not introduce new actions. Define the homogeneous seed

$$H_c = \sum_{\lambda \in \mathcal{R}} x^\lambda e^{N_\lambda L} c_\lambda. \quad (3.6)$$

The desired equation is equivalent to

$$y = H_c + SF(x, y). \quad (3.7)$$

**Theorem 3.2** (Existence, uniqueness, and finite coefficient dependence). *For every input (3.1) and every collection  $(c_\lambda)_{\lambda \in \mathcal{R}}$ , there is a unique solution  $y \in \mathfrak{m}_\Gamma^m$  satisfying (3.2). Every individual block  $P_\gamma$  is obtained by a finite calculation from earlier blocks and finitely many input coefficients. No separation condition on the nonresonant divisors is required.*

*Proof.* At action  $\gamma$ , every occurrence of an unknown block in  $[x^\gamma]F(x, y)$  has action strictly below  $\gamma$ . For a positive-action linear term this follows by subtracting that positive action. For a nonlinear term, at least one other factor has positive action. Constant forcing has no unknown block. Thus well-founded recursion on  $\Gamma_{>0}$ , using Lemma 3.1, constructs a polynomial block at every action. Lemma 2.1 makes each step finite, and the prescribed resonant constants make every step unique.

Equivalently, if  $u, v$  have positive action, expansion of a difference of products gives

$$v(F(x, u) - F(x, v)) \geq v(u - v) + \delta. \quad (3.8)$$

Each linear coefficient has positive action; each nonlinear difference has at least one additional positive-action factor. Since  $S$  does not lower actions, the iteration  $y^{(0)} = 0$ ,  $y^{(n+1)} = H_c + SF(x, y^{(n)})$  has differences whose valuations tend to infinity. Hence each fixed coefficient stabilizes. The support is contained in the already well-ordered set  $\Gamma$ , so the stabilized coefficients define a valid formal sum. The same valuation gain proves uniqueness. Arbitrarily small nonzero numbers  $\gamma - \lambda$  cause no formal difficulty: each individual polynomial calculation divides by only finitely many fixed nonzero numbers.  $\square$

*Remark 3.3* (Which solutions are being classified). The theorem classifies positive-action solutions supported in the chosen  $\Gamma$ . A positive eigenvalue outside  $\Gamma$  does not supply a free homogeneous action in this class. One may enlarge the monoid to include additional positive eigenvalues when appropriate. No assertion is made about solutions with action zero, negative actions, inverse logarithms, or other transseries scales.

## 4 A global degree budget and the exact resonance slope

### 4.1 A coarse budget that closes under nonlinear substitution

For  $\gamma \in \Gamma_{>0}$ , define

$$b(\gamma) = \sum_{\substack{\lambda \in \mathcal{R} \\ \lambda \leq \gamma}} s_\lambda, \quad h(\gamma) = \lfloor \gamma / \delta \rfloor b(\gamma), \quad C = \frac{1}{\delta} \sum_{\lambda \in \mathcal{R}} s_\lambda. \quad (4.1)$$

Set  $h(0) = 0$ . Empty sums are zero. These bounds depend only on the ambient monoid and the constant matrix, not on the coefficients or the chosen resonant constants.

**Lemma 4.1** (Superadditivity and a resonance margin). *The function  $h$  is superadditive on  $\Gamma$ . If  $\alpha + \gamma_1 + \dots + \gamma_n = \lambda \in \mathcal{R}$ , where all  $\gamma_i > 0$ , and the term is allowed by (3.1), then every  $\gamma_i < \lambda$  and*

$$\sum_i h(\gamma_i) \leq h(\lambda) - s_\lambda. \quad (4.2)$$

For  $n = 0$ , the same degree margin holds for constant forcing.

*Proof.* For positive  $u, v$ , monotonicity of  $b$  and superadditivity of the floor give

$$h(u) + h(v) \leq (\lfloor u/\delta \rfloor + \lfloor v/\delta \rfloor) b(u + v) \leq h(u + v).$$

The case involving zero is immediate. At a resonant output action  $\lambda$ , the strict inequality  $\gamma_i < \lambda$  was proved in Theorem 3.2. Consequently  $b(\gamma_i) \leq b(\lambda) - s_\lambda$ , and

$$\sum_i h(\gamma_i) \leq \lfloor \lambda/\delta \rfloor (b(\lambda) - s_\lambda) \leq h(\lambda) - s_\lambda,$$

because  $\lfloor \lambda/\delta \rfloor \geq 1$ . Constant forcing has degree zero, while the last expression is nonnegative.  $\square$

**Proposition 4.2** (Uniform linear degree bound). *The normalized solution satisfies*

$$\ell(P_\gamma) \leq h(\gamma) \leq C\gamma \quad (\gamma \in \Gamma_{>0}). \quad (4.3)$$

*In particular, no resonant action implies a logarithm-free solution.*

*Proof.* Proceed by well-founded induction on the action. At a nonresonant action, the degrees of all products in the source are bounded by the sum of the input degree budgets, hence by  $h(\gamma)$ ; the polynomial solver does not increase degree. At action  $\lambda \in \mathcal{R}$ , Lemma 4.1 bounds the source degree by  $h(\lambda) - s_\lambda$ . The resonant solver increases it by at most  $s_\lambda$ , and the homogeneous seed has degree at most  $s_\lambda - 1 \leq h(\lambda)$ . Other generalized eigenspaces at the same action use nonresonant solvers. This completes the induction. The inequality  $h(\gamma) \leq C\gamma$  follows directly from (4.1).  $\square$

The special resonance margin in (4.2) is important in the analytic proof. A resonant inverse does *not* map every polynomial of degree at most  $h(\lambda)$  back into that same degree space. It maps the smaller, naturally occurring source space into it.

## 4.2 The exact finite-resonance maximum

**Definition 4.3** (Maximal logarithmic slope). For a positive-action vector series with finite block polynomials, put

$$\Theta(y) = \sup_{\gamma \in \Gamma_{>0}} \frac{\ell(P_\gamma)}{\gamma} \in [0, \infty]. \quad (4.4)$$

Zero blocks contribute zero. This is a maximum over all actions when it is attained; it is not a limit superior as the action tends to infinity.

**Theorem 4.4** (Exact resonance control). *For the normalized solution of (3.1),*

$$\Theta(y) = \max \left( \{0\} \cup \left\{ \frac{\ell(P_\lambda)}{\lambda} : \lambda \in \mathcal{R} \right\} \right). \quad (4.5)$$

*Thus the optimal constant in  $\ell(P_\gamma) \leq \theta\gamma$  is attained at a resonant action unless it is zero. The formula uses the actual resonant polynomials, including all cancellations.*

*Proof.* Let  $\theta$  denote the right-hand side, which is finite by Proposition 4.2. We prove  $\ell(P_\gamma) \leq \theta\gamma$  by well-founded induction. At a resonant action this is true by the definition of  $\theta$ . At a nonresonant action, each source product has degree at most

$$\sum_i \ell(P_{\gamma_i}) \leq \theta \sum_i \gamma_i \leq \theta\gamma,$$

since the coefficient action  $\alpha$  is nonnegative. Logarithm-free forcing has degree zero. Sums cannot increase the maximum degree, and the nonresonant solver preserves or lowers it. This proves the upper bound at every action. The reverse inequality follows by taking the supremum over the subset of resonant actions.  $\square$

**Corollary 4.5** (A finite test for logarithm-free solutions). *The normalized solution is logarithm free if and only if  $P_\lambda$  is constant for every  $\lambda \in \mathcal{R}$ .*

*Proof.* All degrees vanish if and only if  $\Theta(y) = 0$ , and Theorem 4.4 characterizes that condition.  $\square$

*Remark 4.6* (What the maximum does not measure). A finite solution  $y = xL$  has  $\Theta(y) = 1$ , even though it has no large-action tail. Hence (4.5) does not claim that  $\deg P_\gamma/\gamma$  approaches the maximum infinitely often. Nor does  $\Theta$  determine a convergence radius or coefficient size. These distinctions are substantive, not just terminological.

## 5 Finite algebraic certificates and cancellation strata

Suppose  $\mathcal{R} \neq \emptyset$ , let  $\Lambda = \max \mathcal{R}$ , and define  $\mathcal{D}$  by (1.3). When  $\mathcal{R} = \emptyset$ , the slope is identically zero and the certificate may be taken empty.

**Theorem 5.1** (Finite input certificate). *Fix  $A$  and  $\Gamma$ . All resonant polynomials  $P_\lambda$ , and therefore  $\Theta(y)$ , depend only on the resonant constants and the finite collection of source coordinates*

$$f_{\alpha,k}, \quad \alpha \in \mathcal{D}, \quad |k| \leq \lfloor \Lambda/\delta \rfloor. \quad (5.1)$$

*Each coefficient of each  $P_\lambda$  is a polynomial in these coordinates and the coordinates of  $c_\lambda$ , with constants depending on the fixed matrix and actions. The resonant polynomials can be computed by recursion over the finite set  $\mathcal{D} \setminus \{0\}$ .*

*Proof.* The set  $\mathcal{D}$  is finite by Lemma 2.3. If a source monomial contributes at action  $\beta \in \mathcal{D}$ , each of its input actions and its coefficient action divides  $\beta$ . Transitivity of divisibility places all of them in  $\mathcal{D}$ . Thus the coefficient recursion on  $\mathcal{D}$  never requires an unknown block outside  $\mathcal{D}$ . The arity bound in Lemma 2.1 is at most  $\lfloor \Lambda/\delta \rfloor$ .

The source calculation uses only additions and multiplications. The polynomial Euler solvers are linear in their source and normalization constant, and divide only by fixed nonzero numbers such as  $\gamma - \lambda$  and fixed positive integers. Induction over the finite ordered set  $\mathcal{D} \setminus \{0\}$  proves polynomial dependence. Including unused coordinates in (5.1) does not change the assertion.  $\square$

**Theorem 5.2** (Algebraic slope strata). *For fixed  $A$ ,  $\Gamma$ , and any real  $q \geq 0$ , the condition  $\Theta(y) \leq q$  is equivalent to the finite set of polynomial equations*

$$[L^d]P_\lambda = 0 \quad \text{for every } \lambda \in \mathcal{R} \text{ and integer } q\lambda < d \leq h(\lambda). \quad (5.2)$$

*In particular, logarithm-free branches are characterized by finite polynomial equations in the source parameters and resonant constants. The possible slope values belong to the finite set*

$$\{0\} \cup \left\{ \frac{d}{\lambda} : \lambda \in \mathcal{R}, 1 \leq d \leq h(\lambda), d \in \mathbb{N} \right\}. \quad (5.3)$$

*Proof.* By Theorem 4.4, the inequality is exactly  $\ell(P_\lambda) \leq q\lambda$  at each resonance. The coarse budget cuts off every polynomial at  $h(\lambda)$ , so this is the finite coefficient-vanishing condition (5.2). Theorem 5.1 makes every displayed condition polynomial in finite input coordinates. The exact maximum is either zero or the ratio of an integer degree to a resonant action, which gives (5.3).  $\square$

The sublevel sets are algebraic in the finite parameter space. Exact slope strata are obtained by subtracting smaller sublevel sets; no claim that they are linear spaces is intended. An arbitrarily large or infinite modification of source coordinates outside (5.1) leaves the slope unchanged. Such a modification can still change convergence, as Section 10 proves.

### 5.1 A finite algorithm when divisor data are available

A practical implementation receives an exact description of  $A$ , the resonances, and the finite divisor set  $\mathcal{D}$ . Sort the positive elements of  $\mathcal{D}$ , compute the allowed source coefficient at each action using already computed blocks, apply (3.4) or (3.5), and retain the resonant blocks. Their coefficients give both the exact slope and the polynomial equations for each stratum. This recursion terminates because  $\mathcal{D}$  is finite, even if infinitely many ambient actions lie below  $\Lambda$ .

There is a necessary distinction between a *finite mathematical certificate* and a *uniform procedure for discovering it*. With arbitrary real exponents or an opaque infinite presentation of  $\Gamma$ , we have not supplied an algorithm deciding membership, equality, or additive divisibility. The finite routine is effective for exact presentations that provide these data, such as a supplied rational prefix known to be divisor closed. It must not silently replace that hypothesis by an arbitrary finite truncation.

## 6 Invariance under logarithm-free coordinate changes

Consider a transformation of dependent variables

$$y = H(x, z) = Bz + G(x, z), \quad B \in \mathrm{GL}_m(\mathbb{C}), \quad (6.1)$$

where  $G$  has logarithm-free coefficients supported in  $\Gamma$ , and its constant and linear terms of action zero vanish. Positive-action translations and linear terms are allowed. No singular factor such as  $x^a$  is permitted in the constant linear part.

**Lemma 6.1** (Formal inverse and slope monotonicity). *The transformation (6.1) has a formal inverse of the same type, with constant linear part  $B^{-1}$ . For every positive-action series  $z$  of finite slope,*

$$\Theta(H(x, z)) \leq \Theta(z). \quad (6.2)$$

*Proof.* For the inverse, solve  $z = B^{-1}y - B^{-1}G(x, z)$  iteratively. Assign weight  $\alpha + \delta|k|$  to a monomial  $x^\alpha y^k$ . Every term beyond the invertible constant linear part has positive additional weight in differences: a positive coefficient action or an additional dependent-variable factor contributes at least  $\delta$ . The same coefficient-stabilization argument as in Theorem 3.2 therefore constructs the inverse. At a fixed coefficient  $(\alpha, k)$ , finite decompositions and the weight bound ensure that only finitely many terms occur. No logarithms are introduced.

Let  $\theta = \Theta(z)$ . A substituted product contributing at action  $\gamma = \alpha + \sum_i \gamma_i$  has logarithmic degree at most  $\theta \sum_i \gamma_i \leq \theta \gamma$ . The coefficient itself has no logarithms. Sums and the constant matrix  $B$  do not increase the bound, proving (6.2).  $\square$

**Theorem 6.2** (Coordinate invariance of the exact slope). *For every transformation (6.1) and every positive-action series  $z$  of finite slope,*

$$\boxed{\Theta(H(x, z)) = \Theta(z)}. \quad (6.3)$$

*In particular, the finite-resonance maximum is a genuine invariant of a normalized solution under this class of local coordinate changes.*

*Proof.* Apply Lemma 6.1 first to  $H$ , and then to its logarithm-free inverse. The first application ensures the transformed series still has finite slope. The second reverses the inequality.  $\square$

For compatible analytic transformations, this is also an invariant of the convergent expansions of the transformed functions. Differentiating (6.1) shows that the transformed differential equation again has a logarithm-free nonlinear remainder and constant linear matrix  $B^{-1}AB$ . The theorem itself, however, only needs the formal transformation and does not require convergence.

The restrictions are essential. A singular shearing  $y = x^a z$  shifts actions and can change every degree-to-action ratio. A change of the independent variable rescales actions. A coordinate change containing logarithms can insert them directly. None of those operations belongs to (6.1).

## 7 A sharp universal criterion for analytic realization

### 7.1 Absolute input convergence and the theorem

Choose a Jordan basis once and use the vector  $\ell^1$ -norm in that basis, with the induced matrix norm. Suppose that, for some  $R, S > 0$ ,

$$\|F\|_{R,S} := \sum_{\alpha \in \Gamma, k \in \mathbb{N}^m} \|f_{\alpha,k}\| R^\alpha S^{|k|} < \infty. \quad (7.1)$$

This is an absolute coefficient condition, not merely a statement that some function has the indicated formal expansion. It gives normally convergent generalized powers in  $x$  and an analytic power series in  $z$  on smaller domains.

For  $P(L) = \sum_d p_d L^d$ , put

$$\|P\|_T = \sum_d \|p_d\| T^d, \quad \|y\|_{r,T} = \sum_{\gamma \in \Gamma_{>0}} r^\gamma \|P_\gamma\|_T, \quad r, T > 0. \quad (7.2)$$

For vector-valued series the second norm is the sum of the component norms. A solution is called *coefficient-absolutely convergent* if  $\|y\|_{r,T} < \infty$  for some  $r, T > 0$ . We will prove that this supplies actual convergent functions on sufficiently small logarithmic sectors.

**Theorem 7.1** (Universal analytic realization). *For fixed  $\Gamma$  and  $A$ , the following are equivalent.*

- (i) *The nonresonant gap  $g(\Gamma, A)$  in (1.4) is positive.*
- (ii) *For every  $F$  satisfying (3.1) and (7.1), and every allowed collection of resonant constants, the normalized formal solution is coefficient-absolutely convergent.*

*If the gap is positive, explicit sufficient norm inequalities for a convergent solution are given in (7.16). If it vanishes, counterexamples can be chosen with no dependence on  $y$ , with all resonant constants zero, and with arbitrarily small input norm. Their formal solutions are logarithm free.*

The universal quantifier is over all coefficient series supported in  $\Gamma$ , not just those supported in a selected proper generating set. The theorem concerns the specific normalized expansion; it does not say that a differential equation has no other analytic interpretation when that expansion diverges.

## 7.2 The degree-budget Banach space

For fixed  $r, T$ , let  $\mathcal{X}_{r,T}$  be the space of series with positive actions, finite norm (7.2), and  $\deg P_\gamma \leq h(\gamma)$ . Zero polynomials are always allowed. It is a Banach space: it is a closed coordinate subspace of the weighted  $\ell^1$ -space indexed by the pairs  $(\gamma, d)$ ,  $0 \leq d \leq h(\gamma)$ . The scalar version is closed under multiplication, by superadditivity of  $h$ , and

$$\|uv\|_{r,T} \leq \|u\|_{r,T} \|v\|_{r,T}. \quad (7.3)$$

We need a slightly smaller source space at resonances. Let  $\mathcal{Q}_{r,T}$  consist of positive-action series  $Q$  with

$$\deg Q_\gamma \leq h(\gamma) \quad (\gamma \notin \mathcal{R}), \quad \deg Q_\lambda \leq h(\lambda) - s_\lambda \quad (\lambda \in \mathcal{R}).$$

These inequalities apply to the full vector polynomial. The source  $F(x, y)$  belongs to  $\mathcal{Q}_{r,T}$  whenever the substitution converges and  $y \in \mathcal{X}_{r,T}$ , by the resonance margin in Lemma 4.1. The same holds for a difference of two such sources. This domain restriction prevents an illicit use of a bounded resonant inverse on a degree space that it does not preserve.

## 7.3 An explicit uniform Euler-inverse bound

Define

$$\Lambda_0 = \max_{\lambda \in \text{spec}(A)} |\lambda|, \quad M = \max_{\lambda \in \text{spec}(A)} \|N_\lambda\|, \quad B_0 = 4(\Lambda_0 + M + 1). \quad (7.4)$$

Choose

$$T \geq \max(1, 8C), \quad D_0 = \lceil CB_0 \rceil. \quad (7.5)$$

Assume  $g = g(\Gamma, A) > 0$ , and set

$$M_0 = \max_{\lambda \in \text{spec}(A)} \sum_{j=0}^{s_\lambda-1} \|N_\lambda\|^j g^{-(j+1)}, \quad (7.6)$$

$$K_{\text{low}} = \sum_{j=0}^{D_0} M_0^{j+1} \frac{D_0!}{(D_0 - j)! T^j}, \quad (7.7)$$

$$K_{\text{res}} = \max \left( \{0\} \cup \left\{ \sum_{j=0}^{s_\lambda-1} \|N_\lambda\|^j \frac{T^{j+1}}{(j+1)!} : \lambda \in \mathcal{R} \right\} \right), \quad (7.8)$$

$$K = \max\{K_{\text{low}}, K_{\text{res}}, 4/B_0\}. \quad (7.9)$$

Powers of a matrix norm with exponent zero are interpreted as one, also when the matrix is zero. These are deliberately conservative constants; they are useful because all dependence is explicit.

**Lemma 7.2** (Bounded right inverse). *For every  $r > 0$ , the zero-normalized polynomial solver defines a bounded linear map*

$$S : \mathcal{Q}_{r,T} \longrightarrow \mathcal{X}_{r,T}, \quad \|SQ\|_{r,T} \leq K \|Q\|_{r,T}.$$

*Proof.* All estimates are coefficientwise in the action. On polynomials of degree at most  $d$ ,

$$\|\partial_L P\|_T \leq \frac{d}{T} \|P\|_T, \quad \|\partial_L^j P\|_T \leq \frac{d!}{(d-j)!T^j} \|P\|_T \quad (0 \leq j \leq d). \quad (7.10)$$

The integrals with zero constant term satisfy

$$\|I_L^k P\|_T \leq \frac{T^k}{k!} \|P\|_T, \quad (7.11)$$

since the coefficient multiplier is  $d!/(d+k)! \leq 1/k!$ .

For a nonresonant block with action  $\gamma < B_0$ , the finite nilpotent inverse has norm

$$\|((\gamma - \lambda)I - N_\lambda)^{-1}\| \leq \sum_{j=0}^{s_\lambda-1} \|N_\lambda\|^j |\gamma - \lambda|^{-(j+1)} \leq M_0.$$

The degree is at most  $C\gamma \leq D_0$ . Inserting (7.10) into (3.4) gives  $K_{\text{low}}$ .

For  $\gamma \geq B_0$ , put  $a = \gamma - \lambda$ . We have  $a \geq 3\gamma/4$ ,  $\|N_\lambda\| \leq \gamma/4$ , and  $\|\partial_L\| \leq h(\gamma)/T \leq \gamma/8$ . Thus, on the finite polynomial space,

$$\|(aI + \partial_L - N_\lambda)^{-1}\| \leq \frac{1}{a - \|\partial_L - N_\lambda\|} \leq \frac{8}{3\gamma} \leq \frac{4}{\gamma} \leq \frac{4}{B_0}.$$

The displayed inverse follows from a norm-convergent Neumann series; it equals the polynomial inverse because that inverse is unique. No resonance occurs at these large actions, since all eigenvalues have absolute value at most  $\Lambda_0 < B_0$ .

At a resonant block, (3.5) with zero constant and (7.11) give  $K_{\text{res}}$ . The source margin makes its output degree at most  $h(\lambda)$ . Other eigenspaces at that action are covered by the nonresonant estimate. In a direct-sum  $\ell^1$ -norm, the maximum of the block bounds bounds the full operator. Finally multiply by  $r^\gamma$  and sum over actions. Absolute summability justifies this last step.  $\square$

The large-action inverse compensates for the growth of polynomial degree; the finite-action gap controls possible small divisors. These are separate parts of the proof. The method is an adaptation of the classical polynomial-norm approach acknowledged in Section 1, rather than a claim that weighted Dulac majorants are themselves new.

## 7.4 Quantitative contraction and sufficiency

For  $0 < r < R$  and  $0 < \eta < S$ , define

$$\Phi_0(r) = \sum_{\alpha > 0} \|f_{\alpha,0}\| r^\alpha, \quad (7.12)$$

$$L_F(r, \eta) = \sum_{\substack{\alpha \in \Gamma, k \in \mathbb{N}^m \\ |k| \geq 1}} |k| \|f_{\alpha,k}\| r^\alpha \eta^{|k|-1}. \quad (7.13)$$

These quantities are finite on smaller radii. In the ball  $\|y\|_{r,T}, \|z\|_{r,T} \leq \eta$ , the usual telescoping product identity and (7.3) imply

$$\|F(x, y) - F(x, z)\|_{r,T} \leq L_F(r, \eta) \|y - z\|_{r,T}. \quad (7.14)$$

The finite homogeneous seed satisfies

$$\|H_c\|_{r,T} \leq \sum_{\lambda \in \mathcal{R}} r^\lambda e^{\|N_\lambda\|T} \|c_\lambda\|. \quad (7.15)$$

**Proposition 7.3** (A usable contraction certificate). *Choose  $T, K$  as above. Suppose  $0 < r < R$ ,  $0 < \eta < S$ , and  $0 \leq \kappa < 1$  satisfy*

$$\boxed{\begin{aligned} KL_F(r, \eta) &\leq \kappa, \\ \|H_c\|_{r,T} + K\Phi_0(r) &\leq (1 - \kappa)\eta. \end{aligned}} \quad (7.16)$$

*Then the normalized solution belongs to  $\mathcal{X}_{r,T}$  and has norm at most  $\eta$ . Such choices exist for every input (7.1) and every fixed collection  $c_\lambda$ .*

*Proof.* On the closed radius- $\eta$  ball in  $\mathcal{X}_{r,T}$ , define  $\mathcal{T}(y) = H_c + SF(x, y)$ . The source lies in  $\mathcal{Q}_{r,T}$ , as proved above. Lemma 7.2 and (7.14) show that  $\mathcal{T}$  has Lipschitz constant at most  $\kappa$ . Moreover

$$\|\mathcal{T}(y)\|_{r,T} \leq \|H_c\|_{r,T} + K\Phi_0(r) + KL_F(r, \eta)\eta \leq \eta.$$

The Banach contraction theorem gives a unique fixed point in this ball. Its coefficient identities and resonant constants are exactly those of Theorem 3.2, so it is the normalized formal solution already constructed.

To choose the radii, first make  $\eta > 0$  small. The action-zero part of  $L_F(r, \eta)$  then tends to zero because it begins at nonlinear degree two. For this fixed  $\eta < S$ , every positive-action part of  $L_F(r, \eta)$  tends to zero as  $r \downarrow 0$ , by dominated convergence using the absolute input norm. The same holds for  $\Phi_0(r)$  and the finite positive-action seed norm. Choose  $r$  small enough to satisfy both inequalities, for example with  $\kappa = 1/2$ . Derivative majorants converge at  $\eta < S$  because  $n(\eta/S)^{n-1}$  is bounded. This verifies all uses of dominated convergence and completes the existence proof.  $\square$

This proves the sufficient direction of Theorem 7.1. Notice that the resonant constants need not be small: their positive actions make the seed small after the radius in  $x$  is reduced.

## 7.5 Necessity and the direction of accumulation

**Lemma 7.4** (Gap failure is left accumulation). *The gap  $g(\Gamma, A)$  is zero if and only if some positive eigenvalue  $\lambda$  is a left accumulation point of  $\Gamma$ . The eigenvalue itself need not belong to  $\Gamma$ .*

*Proof.* Left accumulation gives arbitrarily small nonzero distances and hence a zero gap. Conversely, a sequence of distances tending to zero has, after passage to a subsequence, one fixed eigenvalue because the spectrum is finite. Nonpositive eigenvalues cannot occur: positive actions are bounded below by  $\delta$ . Infinitely many distinct actions approaching the eigenvalue from above would contain a strictly decreasing subsequence, impossible in a well-ordered set. The remaining subsequence approaches from below and may be chosen strictly increasing.  $\square$

*Proof of necessity in Theorem 7.1.* Let  $\lambda > 0$  be a left accumulation point as in the lemma. Choose distinct  $\gamma_n \uparrow \lambda$ , with  $\varepsilon_n = \lambda - \gamma_n \leq 2^{-n}$ , and an eigenvector  $v \neq 0$  satisfying  $Av = \lambda v$ . For any nonzero scalar  $\tau$ , take the source

$$F(x, z) = \tau \sum_{n \geq 1} \frac{\varepsilon_n}{n} x^{\gamma_n} v. \quad (7.17)$$

It is independent of  $z$  and has finite absolute input norm at every fixed  $R, S > 0$ : the actions stay in a bounded positive interval, while  $\sum \varepsilon_n/n$  converges. Its norm can be made arbitrarily small by reducing  $|\tau|$ .

With all resonant constants zero, the coefficient at each  $\gamma_n$  is the constant vector  $-\tau v/n$ , since

$$(\gamma_n I - A)(-\tau v/n) = \tau \varepsilon_n v/n.$$

Thus  $\Theta(y) = 0$ , but for every  $r > 0$ ,

$$\|y\|_{r,T} \geq |\tau| \|v\| \sum_{n \geq 1} \frac{r^{\gamma_n}}{n} = \infty,$$

because  $r^{\gamma_n} \rightarrow r^\lambda > 0$ . The same subseries prevents pointwise absolute convergence at every positive  $x$ . Therefore the universal property fails. This proves necessity and all the counterexample assertions.  $\square$

**Corollary 7.5** (Two independent obstructions). *Finite resonance data determines the exact logarithmic slope without any gap assumption. A zero logarithmic slope does not imply analytic realization. Conversely, a vanishing universal gap does not force every particular input to diverge.*

*Proof.* The first two assertions follow from Theorem 4.4 and (7.17). For the last assertion, the zero source with zero constants gives a convergent zero solution even when the gap vanishes. Section 10 gives a much less degenerate nonlinear family with an exact convergence criterion.  $\square$

## 8 Actual functions and explicit remainder bounds

Let  $y$  have finite coefficient norm  $\|y\|_{r,T} = M_y$ , and suppose its degree bound is  $\ell(P_\gamma) \leq \theta\gamma$ . For a normalized solution one may use the exact  $\theta = \Theta(y)$ , although the coarse constant  $C$  is available without computing the certificate. On a chosen logarithmic cover put

$$q_\theta(x) = \frac{|x|}{r} \max \left( 1, \frac{|\operatorname{Log} x|}{T} \right)^\theta. \quad (8.1)$$

**Theorem 8.1** (Evaluation and action-tail certificate). *On the domain  $q_\theta(x) < 1$ , the series  $\sum_\gamma x^\gamma P_\gamma(\operatorname{Log} x)$  converges absolutely and normally on compact subsets. For every  $B > 0$ ,*

$$\left\| \sum_{\gamma \geq B} x^\gamma P_\gamma(\operatorname{Log} x) \right\| \leq M_y q_\theta(x)^B. \quad (8.2)$$

*The series may be differentiated term by term with  $D = x d/dx$ . When it comes from Proposition 7.3, it solves the actual analytic equation and one may replace  $M_y$  by  $\eta$ . Every sector with bounded argument contains a sufficiently small punctured subsector on which these conclusions hold.*

*Proof.* For a monomial with  $d \leq \theta\gamma$ ,

$$|x|^\gamma |\text{Log } x|^d \leq r^\gamma T^d \left( \frac{|x|}{r} \max(1, |\text{Log } x|/T)^\theta \right)^\gamma.$$

Multiply by  $\|p_{\gamma,d}\|$ , sum over  $d$ , and then sum over actions. Since  $q^\gamma \leq 1$ , absolute convergence follows from the coefficient norm. Restricting to  $\gamma \geq B$  gives  $q^\gamma \leq q^B$ , which proves (8.2). On a compact subset of  $q_\theta < 1$ , the maximum of  $q_\theta$  is some  $q_* < 1$ , so the same majorant proves normal convergence.

For differentiation, the action part contributes a factor  $\gamma$ . The logarithmic derivative contributes at most  $d/T$  to the same weighted monomial majorant, because

$$\frac{|\text{Log } x|^{d-1}}{T^d} \leq \frac{1}{T} \max(1, |\text{Log } x|/T)^d \quad (d \geq 1).$$

Thus the differentiated coefficient is bounded by  $\gamma(1 + \theta/T)q_*^\gamma$  times its coefficient-norm weight. The function  $\gamma q_*^\gamma$  is bounded on  $[\delta, \infty)$ , proving local normal convergence of the derivative. Analytic substitution in  $F$  is permitted by its absolute majorant and  $\|y(x)\| \leq M_y \leq \eta < S$ . Coefficient identities therefore become the analytic differential identity.

Finally, on a bounded-argument sector,  $|x| \max(1, |\text{Log } x|/T)^\theta \rightarrow 0$  as  $x \rightarrow 0$ . This proves the sector assertion.  $\square$

**Corollary 8.2** (Finite asymptotic prefixes). *For a convergent solution as above, any finite initial collection of nonzero action blocks is an asymptotic prefix of the actual function. More precisely, if the next action after that collection is  $\beta$ , the remaining sum is  $O(|x|^\beta \max(1, |\text{Log } x|)^{\theta\beta})$  on a bounded-argument sector. It is therefore smaller than every nonzero logarithmic monomial in the last retained action block.*

*Proof.* Use (8.2) with  $B = \beta$ . The next action is strictly larger than the last retained one, and a positive power of  $|x|$  dominates any fixed power of  $|\text{Log } x|$  as  $x \rightarrow 0$ .  $\square$

*Remark 8.3* (A cutoff need not be a finite computation). The sum over  $\gamma < B$  may itself contain infinitely many blocks. Theorem 8.1 is a bound on an entire action tail, not a promise that ordinary finite truncation has reached that tail. In the accumulating example, reaching action two requires either summing or bounding an infinite block of actions below two. This distinction must be retained in numerical implementations.

## 8.1 A posteriori certification inside the norm ball

The contraction also provides a residual certificate that is useful when coefficients are computed only approximately or whole action blocks are represented by separate analytic routines.

**Proposition 8.4** (Fixed-point residual bound). *Under (7.16), let  $z \in \mathcal{X}_{r,T}$  have norm at most  $\eta$ , and suppose*

$$e \geq \|z - H_c - SF(x, z)\|_{r,T}.$$

*Then*

$$\|y - z\|_{r,T} \leq \frac{e}{1 - \kappa}. \quad (8.3)$$

*For pointwise evaluation, the coarse degree bound gives*

$$\|y(x) - z(x)\| \leq q_C(x)^\delta \frac{e}{1 - \kappa} \quad (q_C(x) < 1).$$

*Proof.* Write  $\mathcal{T} = H_c + SF$ . Since  $y = \mathcal{T}(y)$ ,

$$\|y - z\| \leq \|\mathcal{T}(y) - \mathcal{T}(z)\| + \|\mathcal{T}(z) - z\| \leq \kappa \|y - z\| + e.$$

Rearranging proves (8.3). The difference has positive actions and degree at most  $h(\gamma) \leq C\gamma$ . The evaluation estimate with its smallest possible action  $\delta$  proves the pointwise bound.  $\square$

The word “certificate” here means a theorem with explicitly stated inequalities. A numerical application must actually bound the source majorants and residual; unverified floating-point estimates are not a substitute for those hypotheses.

## 9 Explicit nonlinear examples

### 9.1 An unbounded logarithmic degree with an exact invariant cone

Consider the scalar equation

$$(D - 1)y = y^2 + x(1 + 2y + y^2), \quad \Gamma = \mathbb{N}. \quad (9.1)$$

The logarithm-free change  $z = y/(1 + y)$  has an invertible constant linear part. Direct differentiation gives

$$Dz = \frac{Dy}{(1 + y)^2} = z + x.$$

Thus

$$z = x(L + c), \quad y = \frac{x(L + c)}{1 - x(L + c)} = \sum_{n \geq 1} x^n (L + c)^n. \quad (9.2)$$

The resonant action is one, its normalized constant is  $c$ , and  $\deg P_n = n$  for every  $n \geq 1$ . Therefore  $\Theta(y) = 1 = \Theta(z)$ . The geometric series converges exactly when  $|x(\log x + c)| < 1$ .

This example simultaneously proves three limitations of a naive extension of linear theory. Nonlinear solutions need not have a uniform bound on logarithmic degree. They need not belong to  $\mathbb{C}((x^\Gamma))[L]$ , although they use only one logarithm. And a uniform finite degree bound is not preserved by invertible logarithm-free coordinate changes:  $z$  has degree one, while  $y$  has unbounded degrees. The slope, in contrast, is preserved.

### 9.2 An algebraic locus of logarithm-free branches

Consider

$$Du = u, \quad (D - 2)v = u^2 + ax^2. \quad (9.3)$$

The positive-action normalized solutions are

$$u = cx, \quad v = x^2(d + (c^2 + a)L). \quad (9.4)$$

Consequently

$$\Theta(u, v) = \begin{cases} 0, & c^2 + a = 0, \\ 1/2, & c^2 + a \neq 0. \end{cases}$$

The logarithm-free condition is a genuine nonlinear algebraic constraint on a homogeneous parameter, not a vector-space kernel. Over the reals,  $a > 0$  admits no logarithm-free branch,  $a < 0$  admits two values of  $c$ , and  $a = 0$  forces  $c = 0$ . The parameter  $d$  remains free. Over  $\mathbb{C}$ , the same polynomial equation describes the locus without an order restriction.

### 9.3 A resonant cascade

For  $r \geq 1$ , consider

$$(D - 1)y_1 = x, \quad (D - j)y_j = y_1 y_{j-1} \quad (2 \leq j \leq r), \quad (9.5)$$

with all resonant constants zero. Then

$$y_j = \frac{x^j L^{2j-1}}{(2j-1)!!}. \quad (9.6)$$

For  $j = 1$ , this is  $y_1 = xL$ . If the formula holds for  $j-1$ , the source at action  $j$  is  $L^{2j-2}/(2j-3)!!$ ; a single integration in  $L$  gives (9.6). Hence

$$\Theta(y_1, \dots, y_r) = \max_{1 \leq j \leq r} \frac{2j-1}{j} = 2 - \frac{1}{r}. \quad (9.7)$$

This is a finite cascade of nonlinear resonant integrations. The largest slope is detected at the final resonant action; the calculation does not require estimating an infinite tail.

### 9.4 A Jordan block feeding a nonlinear resonance

Let

$$Du = u + v, \quad Dv = v + x, \quad Dw = 2w + u^2, \quad (9.8)$$

again with zero resonant constants. The solution is

$$v = xL, \quad u = \frac{xL^2}{2}, \quad w = \frac{x^2 L^5}{20}. \quad (9.9)$$

Indeed, the two-dimensional Jordan block integrates the forcing twice across its components, and the last equation integrates  $L^4/4$  at action two. The exact slope is  $5/2$ . The example illustrates why the coarse budget must account for Jordan block sizes, even though the exact theorem only reads off the resulting resonant polynomial degrees.

## 10 An exact nonlinear accumulation threshold

The universal theorem treats all sources on a fixed monoid. A sharper input-specific statement is possible when the accumulation lies below the first nonlinear interaction. It gives a direct nonlinear extension of the linear  $p > 2$  phenomenon recorded in the repository’s Hahn–Fuchsian article [3].

### 10.1 The monoid and the equation

Let

$$E = \{2 - 1/n : n \geq 1\}, \quad \Gamma = \langle E \rangle, \quad \delta = 1. \quad (10.1)$$

Below action two, the only positive actions are those in  $E$ . Every sum of two positive actions is at least two, and equality holds only for  $1 + 1$ . Moreover

$$\text{Div}_\Gamma(2) = \{0, 1, 2\}, \quad \min(\Gamma \cap (2, \infty)) = \frac{5}{2}. \quad (10.2)$$

The final identity follows because the smallest positive generator after one is  $3/2$ : a sum of two generators other than  $1 + 1$  is at least  $5/2$ , and sums of three or more are at least three.

Consider the scalar nonlinear equation

$$(D - 2)y = ax + bx^2 + cy^2 + \sum_{n=2}^{\infty} t_n x^{2-1/n}, \quad \sum_{n=2}^{\infty} |t_n| < \infty, \quad (10.3)$$

with normalization  $P_2(0) = k$ . All four constants  $a, b, c, k$  are arbitrary complex numbers. The input satisfies the absolute coefficient condition at every fixed finite  $x$ -radius and every finite  $y$ -radius, because it is quadratic in  $y$  and the actions of the tail stay in a bounded interval.

**Theorem 10.1** (Finite slope and infinite summability are independent). *The normalized formal solution of (10.3) has*

$$P_1 = -a, \quad (10.4)$$

$$P_{2-1/n} = -nt_n \quad (n \geq 2), \quad (10.5)$$

$$P_2 = k + (b + ca^2)L. \quad (10.6)$$

Its exact slope is

$$\Theta(y) = \begin{cases} 0, & b + ca^2 = 0, \\ 1/2, & b + ca^2 \neq 0. \end{cases} \quad (10.7)$$

Independently of these constants, the normalized solution is coefficient-absolutely convergent if and only if

$$\sum_{n=2}^{\infty} n|t_n| < \infty. \quad (10.8)$$

Thus the entire accumulation tail is invisible to the exact slope, but it decides convergence through a sharp weighted summability test.

*Proof of the coefficient and slope assertions.* There is no quadratic interaction at action below two. At action one,  $(1 - 2)P_1 = a$ , giving (10.4). At action  $2 - 1/n$ ,  $(-1/n + \partial_L)P_{2-1/n} = t_n$ , whose unique polynomial solution is (10.5). At action two, the only product contributing from  $y^2$  is  $P_1^2 = a^2$ ; consequently  $\partial_L P_2 = b + ca^2$ . The normalization gives (10.6). Since two is the only resonant action, Theorem 4.4 gives (10.7). The divisor calculation (10.2) explains the finite certificate directly: none of the actions  $2 - 1/n$ ,  $n \geq 2$ , can be an ancestor of the action-two resonance.  $\square$

*Proof of necessity of the summability condition.* For every  $r > 0$ , the numbers  $r^{2-1/n}$  tend to  $r^2 > 0$ . The constant coefficients in (10.5) therefore have finite weighted norm only when  $\sum n|t_n| < \infty$ . The same reasoning proves failure of pointwise absolute convergence at every positive  $x$  when this sum diverges. Adding a resonant homogeneous term cannot cancel these coefficients at distinct nonresonant actions.  $\square$

*Proof of sufficiency, including nonlinear feedback.* Assume  $\sum n|t_n| < \infty$ , put  $q = b + ca^2$ , and form the entire low-action block

$$Y_0 = -ax - \sum_{n=2}^{\infty} nt_n x^{2-1/n} + x^2(k + qL). \quad (10.9)$$

This is an infinite, but coefficient-absolutely convergent, core. For any fixed  $T > 0$ , its norm tends to zero as  $r \downarrow 0$ . Its degree slope is  $\theta = 0$  when  $q = 0$ , and  $\theta = 1/2$  otherwise. Write

$y = Y_0 + w$ , where  $w$  has actions greater than two. Direct computation using (10.9) reduces the equation to

$$(D - 2)w = c((Y_0 + w)^2 - a^2x^2). \quad (10.10)$$

The right side has no actions at most two: the coefficient of  $Y_0^2$  at two is exactly  $a^2$ , and every other product has action at least  $5/2$ .

Use the coefficient-norm space of series supported in  $\Gamma \cap (2, \infty)$  with  $\deg P_\gamma \leq \lfloor \theta\gamma \rfloor$ . It is complete, and the right side of (10.10) respects this degree bound. On these actions,  $\gamma - 2 \geq 1/2$ . The nonresonant part of the proof of Lemma 7.2, with linear degree constant  $\theta$ , therefore gives a bounded inverse  $S_>$  of  $D - 2$  on this tail space. Let its norm bound be  $K_>$ . For a ball  $\|w\| \leq \eta$ , the map

$$\mathcal{T}_>(w) = cS_>((Y_0 + w)^2 - a^2x^2)$$

has Lipschitz constant at most  $2|c|K_>(\|Y_0\| + \eta)$ , and its value at zero has norm at most  $|c|K_>(\|Y_0\|^2 + |a|^2r^2)$ . First choose  $\eta > 0$  small enough, and then  $r > 0$  small enough, to make the Lipschitz constant less than  $1/2$  and the value at zero at most  $\eta/2$ . When  $c = 0$ , simply take  $w = 0$ . The contraction theorem gives a convergent tail  $w$ .

The resulting  $Y_0 + w$  has the prescribed resonant constant and satisfies the formal equation. By Theorem 3.2, it is the normalized formal solution. This proves sufficiency without assuming a gap below action two; all small divisors there have been absorbed into the summable infinite core.  $\square$

**Corollary 10.2** (The exact power-law threshold). *For  $t_n = \varepsilon n^{-p}$ , with  $\varepsilon \neq 0$  and real  $p > 1$ , the input of (10.3) is coefficient-absolutely convergent and its normalized solution is coefficient-absolutely convergent exactly when  $p > 2$ . This threshold is independent of  $a, b, c, k$ .*

*Proof.* The input requires  $\sum n^{-p} < \infty$ . Theorem 10.1 requires  $\sum n^{1-p} < \infty$  for the solution. These are the stated strict inequalities.  $\square$

## 10.2 Consequences and limitations of the example

Fix  $a, b, c, k$ , and compare the tails  $t_n = \varepsilon n^{-2}$  and  $t_n = \varepsilon n^{-3}$ . They agree in every coordinate of the finite resonance certificate, because that certificate does not contain any  $t_n$ . Their slopes are identical, yet the first normalized expansion diverges and the second converges. Both tails can be arbitrarily small in the input norm. Taking  $b = -ca^2$  makes both formal solutions logarithm free, so absence of logarithms does not weaken the distinction.

The convergence assertion here is stronger than merely applying the universal theorem: the universal gap is zero for this monoid and matrix. The proof instead removes a summable infinite low-action block and uses the positive gap in the remaining tail. This is a precise instance of an infinite-core strategy, not a claim that every accumulating equation admits such a reduction.

When the weighted sum diverges, the theorem excludes the stated coefficient expansion as an absolutely convergent representation. It does not exclude solutions defined by another normalization, a renormalized summation procedure, or direct integration away from zero. Establishing canonical replacements in nonlinear cases is a separate research problem.

## 11 Computation, reproducibility, and a formalization route

### 11.1 What the accompanying program actually checks

The package contains `verification/verify.py` and its recorded `results.json`. The program uses exact SymPy polynomial and rational arithmetic for the finite algebraic checks. Its scalar and Jordan solver uses descending component substitution, providing a second implementation of the operator identities in Lemma 3.1. A finite action dictionary implements coefficient convolution and the normalized nonlinear recursion.

The recorded run used Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0, and random seed 20260929. All checks passed. The categories below are reported separately rather than added into a purported count of independent theorem verifications.

| Check category                                             | Recorded count |
|------------------------------------------------------------|----------------|
| Exact scalar/Jordan block component identities             | 200            |
| Exact nonlinear system coefficient-component identities    | 393            |
| Random coupled systems, included in the preceding category | 16             |
| Random finite-prefix versus resonance slope comparisons    | 16             |
| Geometric and cascade closed-form coefficient comparisons  | 22             |
| Finite prefixes of the accumulating support                | 4              |
| High-precision illustrations of the action-tail bound      | 12             |

The block identities cover Jordan sizes one through four, five nonresonant/resonant gaps (including negative and small rational gaps), and four polynomial degree ranges. Resonant constant terms are checked explicitly. The scalar geometric example is compared through action sixteen; the cascade is checked with six components. The algebraic cancellation example and the Jordan/nonlinear example are checked symbolically. Sixteen random coupled systems are computed through action eight. A forcing perturbation at action five leaves their earlier resonance certificates unchanged, as predicted.

The accumulating example is checked on generator prefixes ending at  $n = 3, 6, 12, 24$ , through action two, with symbolic  $a, b, c, k$ . These are finite prefixes of finite-generator models; they are not a computational proof about the whole infinite monoid. The supplied recurrence is safe only on an action set that is a complete prefix for the finite input, or is divisor closed for the requested targets. Missing an ancestor action invalidates an arbitrary truncation. This precondition is documented in the code.

### 11.2 A numerical illustration with an exact analytic bound

For (9.2) with  $c = 0$ , take  $r = 0.05$ ,  $T = 8$ . Its coefficient norm is exactly

$$M_y = \sum_{n \geq 1} (rT)^n = \frac{2}{3}.$$

After retaining actions through  $N$ , the exact remainder is  $t^{N+1}/(1-t)$ , where  $t = x \log x$ . The theorem bounds its absolute value by  $(2/3)q_1(x)^{N+1}$ . A few recorded values are shown below; mantissas are rounded.

| $x$   | $N$ | Actual absolute remainder | Proved bound            |
|-------|-----|---------------------------|-------------------------|
| 0.001 | 4   | $1.56205 \cdot 10^{-11}$  | $2.13333 \cdot 10^{-9}$ |
| 0.005 | 4   | $1.27111 \cdot 10^{-8}$   | $6.66667 \cdot 10^{-6}$ |
| 0.010 | 8   | $8.90549 \cdot 10^{-13}$  | $3.41333 \cdot 10^{-7}$ |
| 0.030 | 12  | $1.74820 \cdot 10^{-13}$  | $8.70713 \cdot 10^{-4}$ |

The full JSON contains all twelve cases evaluated at 80-digit working precision. These floating-point comparisons illustrate the proved inequality; they are not interval-arithmetic certificates. The conservative coefficient-norm estimate is not advertised as an optimal pointwise bound.

### 11.3 What is, and is not, formalized

The inspected support module names the additive lemma `Fabius.neumann_finite_decompositions` and the corresponding support-preservation bridge [1]. These provide a natural Lean starting point for Lemma 2.3. No new Lean code is included in this package, and the general theorems have not been checked by a proof assistant.

A feasible staged development would first formalize finite additive divisors and divisor-closed recursion. The polynomial solver can then be expressed using a nilpotent matrix, the polynomial derivative, and a zero-constant integration operator. The degree budget and exact slope theorem require only well-founded recursion, finite coefficient convolutions, and degree inequalities. Algebraic parameter dependence can be separated into a finite polynomial-map construction. Finally, the norm estimates and analytic evaluation can be layered on top of this algebraic core using a weighted  $\ell^1$  space and the contraction mapping theorem.

The most important interface decision is to keep *formal support summability*, *coefficient absolute convergence*, and *analytic evaluation* as different structures. The examples in this article show that identifying any two without hypotheses would lose precisely the distinction being studied.

## 12 Further research questions and concrete next targets

The following questions are proposed extensions, not claims that each is an established named open problem. Their value is that they have specific hypotheses, potential obstructions, and test cases.

### 12.1 Tail slope rather than maximal slope

Theorem 4.4 computes the supremum over all actions. Determine when the same value equals  $\limsup_{\gamma \rightarrow \infty} \ell(P_\gamma)/\gamma$ , with an appropriate convention for finite support. The geometric example propagates its resonant slope forever, whereas  $y = xL$  does not. A useful first target is polynomial  $F$  on a finitely generated monoid: characterize persistent maximal-slope propagation through nonlinear dependency paths, including coefficient cancellations. A graph-only criterion will generally need extra noncancellation hypotheses.

### 12.2 The full action–degree region

The scalar  $\Theta$  forgets component information and most of the support geometry. Determine the smallest convex region containing all pairs  $(\gamma, d)$  with nonzero coefficient, or a componentwise

version that records which nonlinear couplings are possible. The degree-budget construction suggests a finite resonance-generated upper region, but equality requires more than an upper-envelope argument. A first task is to compare such regions in triangular systems, where leading logarithmic coefficients can be computed without solving all lower-degree coefficients.

### 12.3 Several accumulation points and weighted divisibility

Theorem 10.1 works because all dangerous actions lie below the first nontrivial quadratic interaction. For several accumulation points, nonlinear sums may land near another resonance. Find input-specific necessary and sufficient weighted coefficient conditions that remain closed under convolution and the Euler inverse. The first genuinely new test case should include two accumulation sequences whose sums approach a third resonant action; otherwise independent scalar weighting may conceal the main issue.

### 12.4 Canonical summation when absolute realization fails

For divergent normalized expansions, construct a canonical renormalized solution class and determine its ambiguity. In the linear repository example an infinite low-action block can be renormalized; nonlinear feedback raises the question whether the renormalization is compatible with products and composition. A precise target is the family (10.3) with  $1 < p \leq 2$ : specify a subtraction prescription, prove that it solves the original analytic equation, and identify the resulting additional asymptotic scales rather than calling an arbitrary regularization a Hahn sum.

### 12.5 Higher logarithmic depth

Replace logarithm-free coefficients by finite iterated-logarithmic blocks and ask for an analogue of the exact resonance certificate. The linear residue and logarithmic-depth infrastructure in ProveIt is relevant, but nonlinear multiplication can propagate degrees across more than one logarithmic level. The first target is a triangular system with coefficients polynomial in  $\log \log(1/x)$  and a positive outer-action gap. One must distinguish a new logarithmic *depth* from unbounded polynomial *degree* at an already present depth.

### 12.6 Nonreal spectrum and oscillatory monomials

The real-spectrum hypothesis lets actions and resonances live in one ordered real monoid. With eigenvalues  $a + ib$ , homogeneous solutions introduce oscillatory factors  $e^{ib \operatorname{Log} x}$ . Develop an ordered support space that separates real growth from imaginary frequency, and determine when finite divisors survive that projection. A finite frequency lattice is a natural first case; arbitrary accumulating frequencies may require a topology rather than only a Hahn order. The analytic gap must then be formulated in the complex spectral plane.

### 12.7 Derivative-dependent nonlinearities and higher order

Our source  $F(x, y)$  contains no derivatives of  $y$ . For equations involving  $Dy, D^2y, \dots$ , differentiation contributes factors comparable to the action and may destroy the bounded nonlinear map used in the contraction. Extend the finite-resonance maximum to such differential polynomials, or construct a counterexample that shows the correct invariant is different. The established

higher-order Dulac convergence theory provides a comparison point; the non-locally-finite support and exact finite certificate remain the specific targets here.

## 12.8 Non-Archimedean actions

In an ordered abelian group of higher rank, positive actions need not have a real lower bound that controls the length of a dependency path. The formal finite-antidiagonal lemma may survive while the bound  $|k| \leq \gamma/\delta$  has no Archimedean meaning. Identify a positive grading into  $\mathbb{R}$  that restores the proof, and describe what happens when no such grading is available. A lexicographic rank-two monoid should be the first test, with the formal recursion and the analytic norm questions treated separately.

## 12.9 Effective discovery of finite certificates

Given a computable description of a support, when can one compute  $\text{Div}_\Gamma(\lambda)$  and certify that it is complete? Theorem 5.1 is not a decision procedure for arbitrary real supports. For restricted differential-algebraic inputs, compare the certificate with the zero-testing framework of Chen, Fang, and van der Hoeven [7], whose 2026 preprint concerns D-algebraic transseries. Arbitrary coefficient families in the present article need not be D-algebraic, so no automatic transfer of decidability is implied. Exact finite-generator and algebraic-action presentations are reasonable initial domains.

## 12.10 Uniformity as resonances move

Allow the eigenvalues and source parameters to vary. For fixed matrix data, the slope sublevels are algebraic; as an eigenvalue crosses a monoid action, the normalization changes and a logarithm can appear. Construct a uniform local chart across such a crossing, with explicit estimates that replace the blow-up of  $g^{-1}$ . An accumulation point of actions meeting a moving eigenvalue is a more singular problem than one isolated crossing. The two should not be treated by the same finite-dimensional normal form without an additional argument.

## 12.11 Certified infinite-block computation

An action-tail estimate is useful computationally only after the retained lower-action block is represented with a certified error. Develop data structures for a finite divisor certificate together with summable infinite blocks, and propagate explicit  $\ell^1$  norm bounds through multiplication, integration, and nonlinear fixed points. The low-action core (10.9) is a concrete benchmark. A successful implementation should report separately its finite algebraic certificate, its block approximation error, and its remaining action-tail bound.

## 12.12 Formal verification of the finite-control theorem

A focused Lean project could target the chain finite divisors  $\rightarrow$  normalized block recursion  $\rightarrow$  exact slope  $\rightarrow$  algebraic strata before tackling analysis. The finite recursion must be stated in a way that does not assume a locally finite action set. The accumulating monoid with  $\text{Div}_\Gamma(2) = \{0, 1, 2\}$  is a useful regression case. A later analytic layer should formalize the restricted source space at resonances; omitting that degree margin would make the claimed bounded right inverse false.

## 13 Conclusion

The nonlinear extension of finite logarithmic control is not a finite global degree bound. It is the exact maximal degree-to-action ratio, attained at a resonance and computed from finitely many divisor coordinates. This invariant survives logarithm-free local coordinate changes and yields algebraic loci of cancellations. The underlying nonlinear solution may nevertheless contain arbitrarily high logarithmic powers.

Convergence answers a different question. A uniform gap to the spectrum is exactly what guarantees analytic realization for every absolutely convergent source on a fixed real Hahn monoid. When the gap fails, finite resonance data does not detect all analytic obstructions. The Riccati family proves that this separation persists under nonlinear feedback: the polynomial  $b + ca^2$  controls logarithms, while  $\sum n|t_n|$  controls convergence. These two certificates—one finite and algebraic, the other potentially infinite and analytic—should remain distinct in both symbolic implementations and future formalizations.

## A Hypothesis and dependency audit

The positive minimum  $\delta$  and real Archimedean action are used to bound nonlinear arity, obtain a uniform valuation gain, and turn a finite set of resonances into a linear degree budget. Well-ordering is used both for recursion and for finite additive divisors. Neither property may be replaced silently by local finiteness, and neither is merely a convergence assumption.

The conditions  $f_{0,0} = f_{0,e_j} = 0$  make the coefficient recurrence strictly triangular in action. A zero-action linear term must first be put in  $A$ . A zero-action constant would leave the positive ideal and is outside the theorem. The absence of logarithms in the source coefficients is used in the exact slope proof: an explicit  $L$ -factor at a nonresonant forcing action could create a new maximal slope there, contradicting the stated formula.

Real spectrum is used to formulate resonant homogeneous modes and small divisors in the same action order. Nilpotent Jordan parts are retained throughout; diagonalizability is not assumed. The global slope uses full vector degrees, so it is insensitive to a fixed invertible choice of basis.

Formal existence and the finite certificate require no analytic input condition and no gap. The analytic theorem requires the absolute coefficient majorant (7.1). The gap is necessary for the *universal* assertion only, and the sharp family deliberately violates it while allowing many convergent inputs. A convergent coefficient norm, not just well-ordered support, is what permits all analytic sums and substitutions in the proof.

The theorem on coordinate invariance is restricted to invertible constant linear parts and logarithm-free coefficients. It does not claim invariance under singular shears, changes of the independent variable, or transformations containing logarithms. The maximal slope is not a tail limit, a pole order, or an exact radius of analyticity.

Finally, the verification program checks finite exact identities and illustrative numerical inequalities. It does not prove the infinite support, Banach-space, or analytic-evaluation theorems. Those rely on the conventional proofs in the article, and no proof-assistant validation or independent peer review is claimed.

## B Repository provenance and inspection scope

The observed repository snapshot used for this article is

3d5973524506411392a911470b5ddc35521568ea

The connected GitHub tools were used to inspect the transseries directory structure, the documentation index, the Hahn–Fuchsian and residue-obstruction package descriptions, a portion of the Hahn–Fuchsian source, and the complete support module named in the bibliography. The large documentation index and the Hahn–Fuchsian source exceeded the response display budget; no complete line-by-line audit of those large files, the canonical volume, or the whole repository is claimed. Newly delivered archives at the repository’s intake were not exhaustively audited.

The source-level support bridge was read at the pinned snapshot. The mathematical theorems in the present article are proved self-containedly rather than assumed from an unreviewed repository report. External literature was consulted through primary author, journal/preprint metadata, and paper text. The Dulac paper’s PDF text was accessible, but the remote screenshot endpoint failed; no result here relies on an unseen figure or graphical measurement. The PDF of this article was separately rendered and visually checked in the local build workflow.

No repository files were modified. A suitable proposed placement is `Analysis/Transseries/docs/series-and-transseries/Nonlinear_Hahn_Dulac/`. This is a packaging suggestion, not an upload that has been performed. The archive is standalone and does not load the repository’s large canonical volume or notation files.

## C Reproduction instructions

The archive contains the editable `nonlinear_hahn_dulac.tex`, the compiled PDF, `build.sh`, the verification program and recorded JSON, pinned verification requirements, and separate proof/provenance/build notes. The bibliography is embedded in the LaTeX source; no bibliography processor is needed.

To rebuild the article with a standard TeX Live installation, run

```
sh build.sh
```

To repeat the finite verification, run

```
python -m pip install -r verification/requirements.txt
python verification/verify.py
```

The verification program uses no network access after its dependencies are installed. Random tests use a fixed seed. A successful run writes `verification/results.json`. Exact algebraic assertions raise an exception on failure; the program must be run without Python’s `-O` option, which would disable assertions.

## References

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